

The Unequal Visibility of Epidemic Death: Smallpox at the Cape, 1713*

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Abstract

Unequal death recording can obscure the social distribution of epidemic mortality. Around the Cape’s 1713 smallpox epidemic, deaths are documented for 22.2 per cent of men with five or more adult slaves recorded, against 6.6 per cent of men with none. A reconstruction links 348 men named in the 1712 tax roll to estate documents and widow entries, confirming 33 all-cause deaths through 1714. Genealogical presence and a comparison restricted to men with a wife named show why family circumstances matter to interpreting this gap. A recording model establishes that equal mortality would require higher-resource deaths to be about 3.4 times as likely to be recovered. The sources do not estimate that recovery difference. The contribution is a reviewed death register and an explicit benchmark for assessing mortality comparisons when property and family relationships shape both survival and its documentation.

Keywords: epidemic mortality; record linkage; historical demography; archival selection; partial identification; smallpox; Cape Colony

JEL codes: N37; I14; J11

*I thank Jesse Naidoo for detailed comments on earlier drafts and the LEAP research group at Stellenbosch University. I thank Helena Liebenberg for sharing the Tracing History Trust’s dagregister transcription, and acknowledge the Cape of Good Hope Panel and TEPC transcribers. An online appendix with source decisions, model derivations, sensitivity checks and record-continuation diagnostics accompanies this paper. This paper was created with the help of Anthropic’s Claude Code (Opus 4.8 and 5.5) and OpenAI’s Codex (GPT-5.5, GPT-5.6 and GPT-6). Cite this paper as: Fourie, Johan. 2026. “The Unequal Visibility of Epidemic Death: Smallpox at the Cape, 1713.” Working Paper, Department of Economics, Stellenbosch University.

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1 Introduction

Around the Cape Colony’s 1713 smallpox epidemic, deaths are documented for 22.2 per cent of men with five or more adult slaves recorded, compared with 6.6 per cent of men with none. These figures represent 10 of 45 and 16 of 243 men named in the Stellenbosch–Drakenstein tax roll of 1712. They count confirmed deaths from all causes after that enumeration through the end of 1714. Why are deaths so much more frequently documented among men with greater recorded resources? Does this reflect mortality, recording, or both?

This research note reconstructs 33 confirmed deaths among 348 individually named men and examines the institutions through which their deaths become visible. Estate papers and widow entries confirm deaths; genealogical records show broader differences in family representation. A recording model then quantifies the recovery difference required to explain the observed comparison under equal mortality. The contribution is a reviewed death register and a benchmark against which further evidence about recording can be assessed.

Resources could help a household through an epidemic. When adults and children fall ill together, survival may depend on someone remaining able to provide food, water and nursing (Schneider and Davenport 2026). Yet wealth has not offered the same protection in every period or against every disease (Bengtsson and Dribe 2011; Alfani 2022; Dribe and Karlsson 2022). Age, prior exposure and household composition may differ across resource groups. Higher mortality among men with greater resources remains one possible explanation of the recorded difference.

The same property and family relationships that could support survival also created routes into estate administration and later records. Research on epidemic costs emphasises households’ resources and institutional circumstances (Curtis, van Besouw and Hilken 2025); these circumstances also matter when reconstructing who died. If deaths were equally likely to be recovered across groups, their recorded-death ratio would preserve their mortality ratio despite incomplete coverage. Only complete recovery would make the recorded percentages themselves mortality rates.

The preceding tax roll supplies names and resources before the outbreak; estate papers and widow entries provide different routes to confirming deaths. This combination defines a baseline population independently of the documents generated by deaths.

Historical demographers have long treated source verification as part of demographic inference (Henry 1968; Alter 2020). Family reconstitution selects people whose events can be connected, and migration or missing events can alter the histories recovered (Ruggles 1992; Wrigley 1994; Desjardins 1995; Kasakoff and Adams 1995). Linkage research similarly distinguishes erroneous matches from differences in who can be linked (Ruggles, Fitch and Roberts 2018; Bailey et al. 2020; Abramitzky et al. 2021). The Cape reconstruction connects these concerns to a concrete mortality comparison: the low-to-high recorded-death ratio of 0.30 is consistent with equal mortality if higher-resource deaths were about 3.4 times as likely to be recovered. This is a threshold, not an estimate of the actual recording difference. The paper examines the records and family circumstances before stating the mortality comparisons consistent with them.

2 The epidemic and the records

2.1 Smallpox at the Cape

The Dutch East India Company established its Cape refreshment station in 1652. By the early eighteenth century, settlers occupied farms across Stellenbosch and Drakenstein, with large differences in their command of land, livestock and enslaved labour. These administrative records have supported research on wealth, mobility and family formation (Fourie 2013; Fourie et al. 2025; Cilliers et al. 2025). They describe settlers much more fully than the Khoen and San, whose land and livelihoods were transformed by conquest and dispossession (Marks 1972; Guelke and Shell 1992; Adhikari 2010). Accounts of the 1713 epidemic describe disruption among Khoi communities, although reconstructions differ over the magnitude and timing of population losses (Fourie and Green 2015; La Croix 2018; Viljoen 2023). Enslaved people usually appear as counts attached to a roll entry. These records cannot supply a corresponding individual analysis of enslaved, Khoen or San deaths, and the named male cohort does not represent the colony’s whole population.

The Company’s daily register establishes the outbreak’s progression. It records increasing smallpox among Company slaves on 31 March 1713, rising deaths in April and spread towards the farming districts in May. In June the register reports severe illness in Drakenstein; the entry of 28 November describes continuing cases and labour shortages. An entry of 6 May identifies people born locally, and others who had not previously experienced smallpox, as especially exposed.¹ Some adults arriving from Europe could therefore have acquired immunity before reaching the Cape.

Smallpox maintained by repeated transmission in a large population often affected children; a long interval without an outbreak allowed susceptibility to accumulate across ages (Duncan, Scott and Duncan 1993). Simultaneous illness could leave a household without anyone able to fetch water, prepare food or nurse its members (Schneider and Davenport 2026). Resources might ease that disruption, but age, prior exposure and family composition could also differ across resource groups. The comparison below is descriptive and does not estimate the causal effect of wealth on mortality.

2.2 Tax rolls, estate documents and family records

The *opgaafrolle* recorded names, family members, resident labour, enslaved people, livestock and production for the Company’s administration. The Stellenbosch–Drakenstein series begins in 1685 and forms part of the Cape of Good Hope Panel (Fourie and Green 2018; Fourie et al. 2024). An entry is not invariably a separate household: bachelor sons, employees and other associated people could receive their own rows. Resources recorded against such an entry need not measure the resources of the household in which the man lived. A male-name field can also contain a joint entry or an agent acting for an estate. I return to the original 1712 transcription to define individually

¹VOC Cape daily register, entries for 31 March, 18 April, 6 and 10 May, 13 June, 24 June and 28 November 1713, in the Tracing History Trust transcription shared by Helena Liebenberg. The replication files retain these seven entries with their dates.

named men and their recorded resources.

The transcription’s missing-data rule is documented. When a variable has an entry somewhere in a census, blank cells in that column are coded as zero; an entirely empty column denotes a variable not recorded in that census (Fourie et al. 2024, p. 435). Both adult-slave columns and each livestock column used here contain positive entries in both districts in 1712. I apply the documented rule to those columns. “No adult slaves recorded” consequently has a specific meaning, although self-reporting and under-reporting still separate recorded holdings from actual resources. The category does not imply an absence of property or identify all poor residents.

The Orphan Chamber’s estate documents, catalogued as MOOC8, supply the first route to confirming deaths. They include inventories, accounts and distributions. The Chamber’s responsibilities towards minors, absent heirs and estates made probate selective (Fourie 2013). Some deaths produced several documents, while others produced none. Nor is the first person named in a document necessarily dead: a surviving husband or widow may declare the household’s property. I count documents in the annual series and individually confirmed deaths in the linked analysis.

Widow entries in later tax rolls supply a second route. A woman may be named as a particular husband’s widow or identified through her own name. Connecting her to a baseline couple can establish that the husband died. This requires attention to family sequence: a widow already present before 1712 supplies no new death for the cohort, and a living son may share his deceased father’s name. Continued enumeration also depends on remarriage, migration, resources and household organisation. Widow entries extend coverage beyond estate papers, but a man without a surviving, identifiable widow cannot be recovered through this route.

The South African Families database (SAF) reconstructs settler families from church registers and other genealogical sources (Cilliers and Mariotti 2019; Cilliers 2021). Membership of a reconstructed lineage provides evidence about identity and historical representation. It does not establish death during the observation interval. I report existing SAF linkage alongside the death channels and use family evidence to distinguish namesakes where the underlying sources can be checked. Such differences in observation are familiar from models of passive registration and comparisons with active population registers (Jonker and van der Vaart 2007; Alter, Devos and Kvetko 2009; van den Berg et al. 2021). Here the family is both a source of identifying information and a route through which death can become documented.

2.3 The epidemic in the administrative series

Both administrative series show a sharp disturbance in 1713 (Figure 1). The MOOC8 collection contains 54 documents dated that year, compared with a mean of 14.5 in the three preceding and three following years, a ratio of 3.7. In the district rolls, 21 of 305 enumerated entries are widows, a share of 6.9 per cent. The pooled share over 1708–12 is 2.6 per cent.

These series use no person linkage. Their agreement with the daily register locates the mortality shock, but their magnitudes describe administrative activity and the stock of widow entries. Several documents can arise from one death, and widow shares change with widowhood, remarriage,

migration and enumeration. Neither series supplies a population mortality rate.

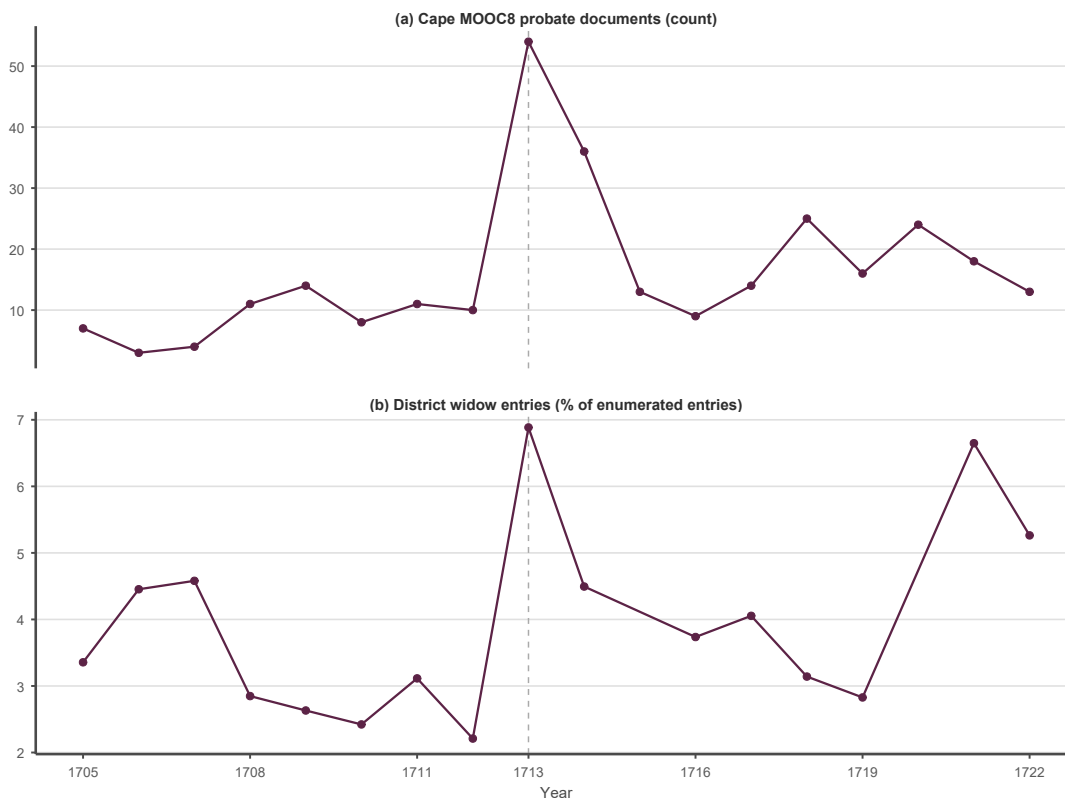


Figure 1: Probate documents and widow entries around the 1713 epidemic. The upper panel counts dated MOOC8 documents across the Cape; the lower panel gives widow entries as a percentage of district entries with a male name or widow status. Points denote observed years; missing rolls are not zero counts. The dashed line marks 1713.

2.4 The cohort and the death register

The raw 1712 transcription contains 354 male-name entries, from which I retain 348 individually named men. Six excluded rows comprise a joint entry, an estate representative and two unresolved repeated-name pairs. The review retains source-row identifiers and uses full names, wives and family sequences to distinguish people. Online Appendix A sets out the decisions, including cases in which similar names denote different men. Unresolved pairs enter a separate denominator sensitivity check.

Adult slaveholding supplies the primary resource grouping: none recorded, one to four, and five or more (Table 1). The lowest group contains 69.8 per cent of the cohort. An alternative index combines adult slaves with cattle, horses and sheep. Measuring resources before the outbreak avoids using asset changes caused by subsequent illness to define the groups, while retaining the distinction between a man's recorded resources and an independent household's total resources.

Death classification follows a separate source review. I examine all 90 MOOC8 documents dated

Table 1: The 1712 cohort by recorded adult slaveholding

Adult slaves	Men	Share (%)	Mean slaves	Mean cattle	Mean sheep	Wife named
0 recorded	243	69.8	0.0	5.1	34	74
1–4	60	17.2	2.2	24.7	273	48
5+	45	12.9	8.8	67.4	708	41

Notes: Individually named men after source review. A row need not represent an independent household. Blank asset cells follow the transcription’s documented zero-coding rule. Cattle include cows and oxen. Wife named is a count of entries with a recorded wife, not an estimate of the number married.

1713–14 and all 33 widow entries in the two district rolls, including statements within document bodies. An accepted link requires a deceased person or a widow’s husband who can be connected to a baseline man. Names, spouses, children and earlier widow entries help establish identity. Explicit references to a deceased man in someone else’s estate papers also count as death evidence; an appendix check excludes these incidental references.

The resulting register confirms 33 cohort deaths by the end of 1714. Repeated probate documents and overlapping widow evidence count once per man. Unresolved widow assignments enter separate sensitivity calculations. A document dated in 1714 gives an upper date for a death, not an automatic assignment to the epidemic months of 1713. Men without an accepted death link remain unclassified as to survival.

3 Recorded deaths and family circumstances

3.1 What the different sources record

Table 2 brings the opening comparison back to its sources. In the highest-resource group, 10 of 45 men have a probate or widow confirmation, compared with 16 of 243 in the lowest group. The corresponding rates are 22.2 and 6.6 per cent; the middle group has 7 confirmed deaths among 60 men, or 11.7 per cent. The combined measure counts each deceased man once even when both sources confirm his death.

The two death channels have different coverage but the same ordering. Estate documents confirm 5.8 per cent of the lowest group and 17.8 per cent of the highest. Widow entries confirm 1.6 and 11.1 per cent, respectively. This agreement does not separate recording from mortality: each channel needs a death as well as a surviving record. It does show why adding widow evidence to probate cannot be assumed to make the combined register socially representative.

Existing genealogical linkage rises from 3.7 to 26.7 per cent. Unlike the first two columns, this is not an outcome requiring death during the interval. It shows that the contrast in death records accompanies a broader difference in which men appear in reconstructed families. But genealogical membership does not measure the probability of recovering a death and cannot calibrate the missing deaths in either group. Online Appendix C plots the source-specific rates and their intervals.

Table 2: Confirmed death records and genealogical presence

Adult slaves recorded	MOOC8 death confirmation	Widow entry	Combined death record	SAF presence
	14/243	4/243	16/243	9/243
0 recorded	5.8 [3.5, 9.4]	1.6 [0.6, 4.2]	6.6 [4.1, 10.4]	3.7 [2.0, 6.9]
	5/60	2/60	7/60	9/60
1–4	8.3 [3.6, 18.1]	3.3 [0.9, 11.4]	11.7 [5.8, 22.2]	15.0 [8.1, 26.1]
	8/45	5/45	10/45	12/45
5+	17.8 [9.3, 31.3]	11.1 [4.8, 23.5]	22.2 [12.5, 36.3]	26.7 [16.0, 41.0]

Notes: Each cell reports the count/denominator above the percentage and its 95 per cent Wilson interval. MOOC8 confirmation includes explicit death mentions in other estates; repeated documents count once per man. Combined death record counts each man once across the first two columns. SAF presence is an existing genealogy link and does not signify a death. Intervals describe binomial reference-model uncertainty conditional on the cohort and accepted links; they do not cover unobserved deaths or uncertain identity decisions.

3.2 The role of family circumstances

A wife is named for 74 of the 243 men in the lowest group, compared with 41 of 45 in the highest. This difference matters directly for a death channel that follows a surviving widow. Among men with a wife named, the combined recorded-death rates are 14.9 per cent (11/74) and 22.0 per cent (9/41). The comparison is therefore much closer than the headline rates of 6.6 and 22.2 per cent.

The restriction holds one aspect of family documentation more nearly constant. It also selects different populations: separately listed sons or employees may differ from established family men in age, prior exposure and access to household care. A blank wife’s name does not establish that a man was unmarried. The companion reconstruction traces inheritance and remarriage after bereavement (Fourie 2026). In Johannes Viellion’s estate, relatives and a church official arranged an inventory to protect a surviving child’s interests.² Such relationships helped generate evidence of death as well as organise its consequences. The narrower gap consequently cannot be attributed wholly to recording, nor interpreted as an adjusted mortality comparison. It shows why the social composition of the baseline entries and their eligibility for later records belong in the explanation of the headline difference.

Alternative resource definitions and source decisions preserve the ordering of recorded deaths. Unresolved widow assignments yield low-to-high ratios of 0.269–0.315; they do not resolve the mortality ranking. Online Appendix C reports these checks, including the broader resource index and shorter documentary search.

4 From recorded deaths to mortality

The probability of recovering a confirmed death combines the probability of dying with the probability that the death leaves evidence which survives and can be linked. Let M_w be the probability

²TEPC transcription, Orphan Chamber, MOOC8/3.49, 4 July 1713.

that a man in resource group w dies after the 1712 enumeration and by the end of 1714. Let C_w be the probability of recovering such a death under the review protocol, conditional on its occurrence. These are cumulative probabilities over the observation interval, not annual mortality rates. The conditional recovery probability is the model’s capture probability. If accepted records identify the correct person and interval, the recorded-death probability is

$$D_w = M_w C_w. \quad (1)$$

A recorded rate of five per cent could therefore arise if five per cent died and all deaths were recovered, or if ten per cent died and half were recovered. Better linkage helps establish the recorded rate. Estimating mortality also requires information about the deaths that do not appear.

Write L for men with no adult slaves recorded and H for those with five or more. Their mortality ratio is

$$\theta \equiv \frac{M_L}{M_H} = \frac{D_L/D_H}{C_L/C_H} = \frac{D_L/D_H}{\kappa}. \quad (2)$$

The observed low-to-high recorded-death ratio is 0.296, with an approximate 95 per cent interval of 0.14–0.61 under a binomial reference model. That interval describes uncertainty in the recorded comparison conditional on the cohort and accepted links. It does not include uncertainty about lost records or historical identities. The additional unknown in equation (2) is κ , the relative probability of recovering a lower-resource death.

Equal mortality requires $\kappa = D_L/D_H$: deaths in the higher-resource group would need to be about 3.4 times as likely to be recovered. This threshold quantifies the recording difference that would account for the entire observed gap. It does not establish that such a difference existed. If lower-resource deaths were half as likely to be recovered, their implied mortality ratio would be 0.59; if a quarter as likely, it would be 1.19. Figure 2 shows where the ranking changes, using the combined death record. These are conditional calculations rather than estimates of recording completeness.

The institutions give reasons to question equal recovery. Property gives the Orphan Chamber an estate to administer, and surviving family members can create later records and help establish identity. Yet probate also depends on heirs and legal circumstances; widow evidence requires a surviving wife. Table 2 cannot quantify these conditional probabilities because the true deaths supplying their denominators are unknown. Even a higher recovery probability in each source separately would not establish higher combined recovery without considering overlap between sources.

Suppose, as an explicit assumption, that higher-resource deaths were at least as likely to be recovered in the combined record: $C_L \leq C_H$. This raises the lower bound on the mortality ratio to the observed recorded-death ratio. The sharp range consistent with the observed rates and this assumption is [0.296, 4.50]. It includes equal mortality and a mortality disadvantage in either direction. Online Appendix B derives the bound and distinguishes it from sampling uncertainty. Institutional knowledge narrows the possible comparisons but does not select the ranking.

Later name records supply no independent resolution. Disappearance can reflect migration,

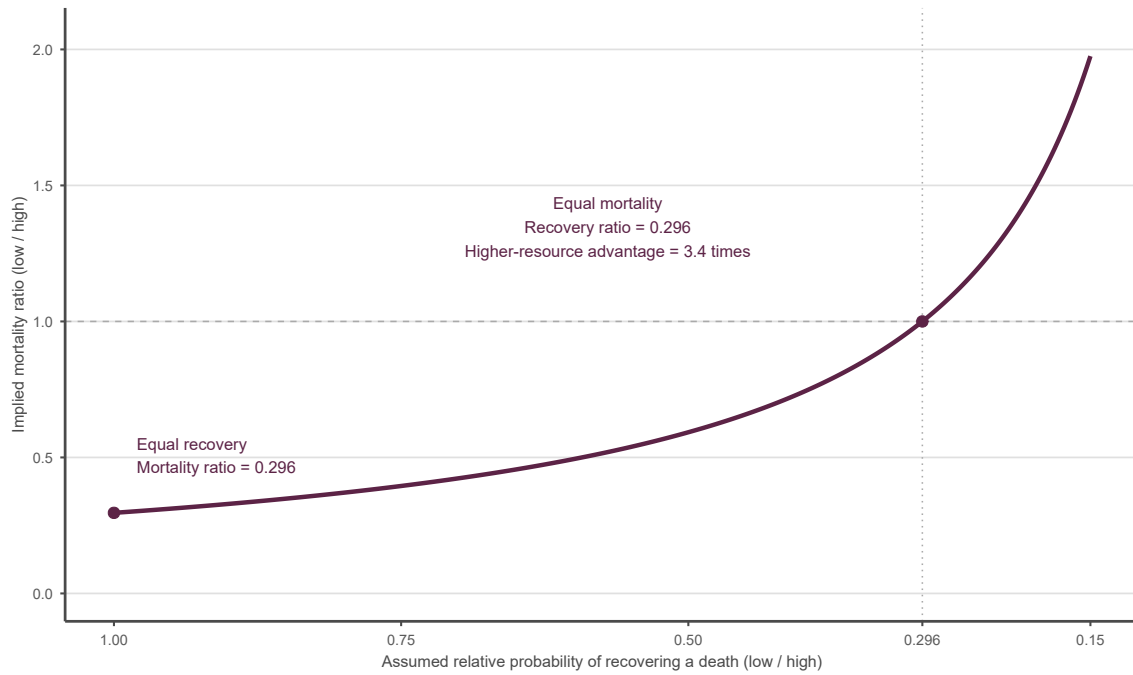


Figure 2: How unequal recording changes the mortality comparison. The curve uses the combined probate and widow death record. The horizontal axis is the assumed probability of recovering a lower-resource death relative to a higher-resource death, C_L/C_H ; the vertical axis is the implied mortality ratio, M_L/M_H . Equal recovery preserves the observed ratio of 0.296. Equal mortality occurs at a recovery ratio of 0.296, equivalent to a higher-resource advantage of about 3.4. The curve does not estimate recovery probabilities or include uncertainty about missing records.

naming or enumeration changes as well as death. The imprecise comparison of continuation rules in Online Appendix D neither identifies the mortality ranking nor validates missing names as death proxies.

5 Discussion and conclusion

Deaths were more frequently documented among men with greater recorded resources. The reconstruction establishes this gradient in recoverable deaths without identifying a corresponding gradient in smallpox mortality. Reading documentary roles and family sequences prevents living declarants from being counted as dead and recovers deaths missed by searches confined to document headings. Restricting the comparison to men with a wife named narrows the gap, showing why family documentation and the composition of the baseline population matter together.

The economic-demographic problem is that resources and family circumstances enter both the survival process and the production of evidence. Property could support care but also bring an estate within the Orphan Chamber’s administration; a surviving spouse could provide assistance and later supply a widow record. The evidence establishes routes through which those processes could operate, not their separate effects. A household-care explanation remains plausible, but these data cannot establish whether it produced higher mortality among men with fewer resources.

The recording model makes the remaining uncertainty quantitative. Equal mortality would require higher-resource deaths to have been about 3.4 times as likely to be recovered. This benchmark states what a recording explanation must account for. The sources do not locate the actual recovery difference relative to it, and the assumption that higher-resource deaths were at least as recoverable still permits either mortality ranking.

Further progress requires evidence about coverage: independently enumerated deaths, estate settlements outside the Orphan Chamber, or surviving spouses whose records establish eligibility for widow registration. Additional overlapping sources can improve the register without revealing how many deaths every source missed. The wider contribution is therefore a way to connect a defensible baseline population, reviewed death evidence and institutional knowledge to an explicit demographic comparison. Other historical settings can use the same steps, while requiring their own assessment of coverage and recovery probabilities.

Data and computational assistance

Replication files contain source-row identifiers, death-link decisions and code; transcription access follows custodians’ conditions. Claude Code (Anthropic; Opus 4.8, 5.5) and Codex (OpenAI; GPT-5.5, 5.6, 6) assisted with code, source comparison, literature review, mathematical checks and revision. Saved scripts produced all numerical results. AI-assisted historical judgments are not independent human validation.

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